Periodic Table SME Agent

An Autonomous Retrieval-Augmented Reasoning System

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Overview

1. Data Preparation & Ingestion
2. Embedding, Indexing & Retrieval
3. Agentic Workflow (LangGraph)
4. System Implementation
5. Experiments & Results
6. Bonus Implementations

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Data Collection Curation

Strategy: High-quality, domain-specific curation over raw quantity.

- **Primary Source: NCERT Chemistry Textbooks**
  - **Manual Traversal:** We manually reviewed chapters across Grades 9-12.
  - **Relevance Grading:** Each chapter was manually graded on its relevance to "Periodic Table" concepts. Only high-relevance chapters were admitted to the corpus.
- **Supplemental Sources:**
  - Integrated external educational materials from platforms like **Vedantu** and **Byju's**.
  - **Purpose:** To provide diverse explanations, solved examples, and alternative problem sets not found in standard textbooks.

Outcome

A focused, noise-free dataset optimized for educational accuracy.

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Data Ingestion Pipeline

Objective: Robust handling of heterogeneous formats.

Supported Formats:

- **PDF:** Layout-aware extraction via `pdfplumber`.
- **DOCX/PPTX:** Paragraph/shape extraction.
- **Structured:** CSV with dialect sniffing.

Preprocessing:

- **Cleaning:** Regex for artifacts.
- **Normalization:** Whitespace collapse.
- **Deduplication:** SHA256 hashing.

Bonus (OCR Fallback): Automatically triggers Tesseract OCR if standard extraction yields < 20 characters/page.

Hierarchical Chunking Strategy

Addressing "Lost in the Middle" via Parent-Child relationships.

- **Method:** Content-Aware Paragraph Packing.

Level	Tokens	Purpose
Parent	2048	Broad Context
Child	512	Standard Context
Leaf	128	Granular Facts

Metadata links Leaf $\rightarrow$ Child $\rightarrow$ Parent for context expansion.

Embedding Indexing

Manual Embedding Pipeline:

- **Model:** Agnostic transformer loading.
- **Pooling:** Mean Pooling + Attention Masking + L2 Normalization.

Elasticsearch Schema:

- `dense_vector` (Cosine Similarity)
- `text` (BM25 Keyword Search) for **Hybrid Retrieval**.

Retrieval Architecture (RAG Flow)

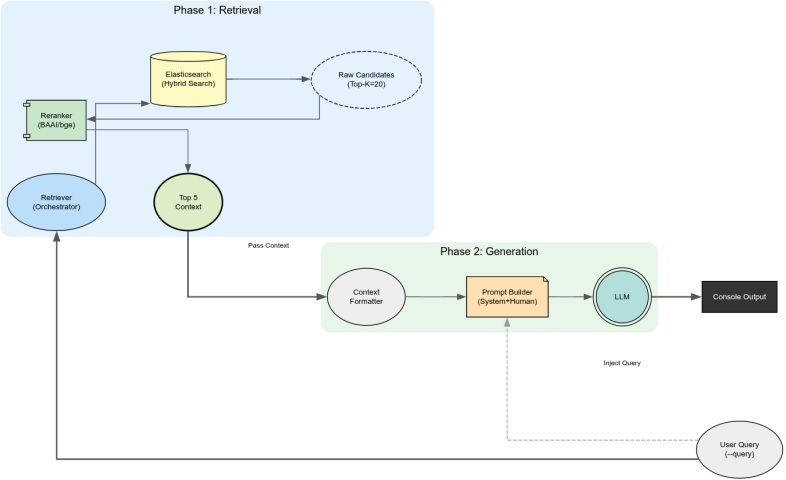

Figure: Visualizing the Hybrid Retrieval and Reranking Pipeline

Retrieval Strategy Details

1. **Hybrid Search:** Combines Vector k-NN + BM25 Keyword matching.
2. **Semantic Reranking (Bonus):** Top-20 candidates re-scored using a Cross-Encoder (BGE-Reranker).
3. **"Small-to-Big" Expansion:** Match on dense 128-token chunks $\rightarrow$ Return 2048-token parent context.

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Agent Architecture

Core Logic: LangGraph

- **State Machine:** Tracks tasks and content.
- **Cyclic Dispatcher:** Routes between Report, Quiz, and Email generation.
- **Self-Correction:** Injects dependencies (e.g., generate report before presentation).

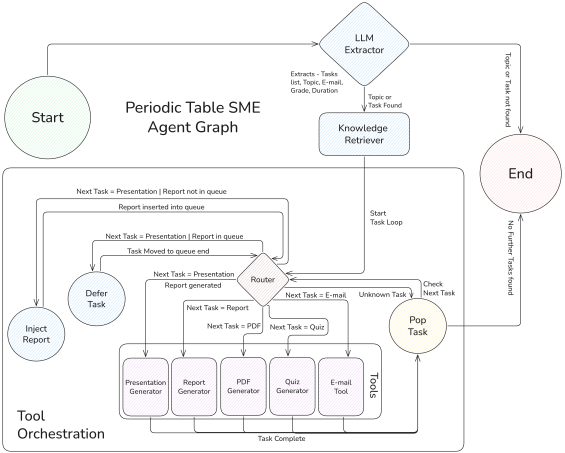

Figure: LangGraph Orchestration Tool Routing

Tool Capabilities

- **Knowledge Retrieval:** JIT Model offloading to prevent OOM.
- **Document Generation:**
  - `markdown_pdf`: Clean, professional reports.
  - `python-pptx`: Automated slide decks with custom styling.
- **Communication:** SMTP Email with Markdown-to-HTML rendering.

System Interaction API Structure

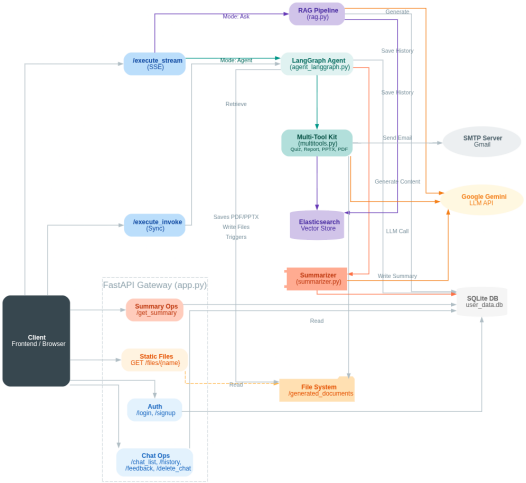

Figure: Full Stack Architecture: FastAPI, SSE, and Database Interactions

Implementation Highlights

- **Backend (FastAPI):** Async architecture with Server-Sent Events (SSE) for real-time streaming.
- **Persistence:** SQLite with separated `chat_summaries` and `user_summaries`.
- **Memory Optimization:**
  - **Summarizer Module:** Compresses conversation history using Gemini Flash Lite.
  - Reduces context window usage while maintaining long-term memory.

Quantitative Analysis

Dataset: Curated Chemistry QA Test Set ($N = 135$).

Model	Accuracy	Precision	Recall	F1-Score
Gemini 2.5 Flash	**0.811**	**0.905**	**0.811**	**0.824**
Gemini 2.5 Flash Lite	0.730	0.850	0.730	0.753
Qwen 1.7B (8-bit)	0.642	0.701	0.617	0.653

Table: Performance Comparison

- **Flash:** Superior reasoning.
- **Qwen (Local):** Viable for direct fact retrieval.

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Bonus Features Summary

1. **Automated Batch Ingestion:** Multiprocessing pool reduces I/O bottlenecks.
2. **Relevance Reranking:** Cross-Encoder for precision scoring.
3. **Feedback Loop:** Adapts agent planning based on historical user critique.
4. **Security Guardrails:** XSS prevention and JSON injection defense.

Thank You

Team Aurors

Questions?